

- Find distance from pitch line to back B from the table

Diametral pitch	Pitch line to back (B)	Overall thickness	Face width	Nominal length [ft]
64	0.109	0.125	0.125	2
48	0.104	0.125	0.125	2
32	0.156	0.187	0.187	4
24	0.208	0.250	0.25	4
20	0.450	0.500	0.5	6
16	0.688	0.750	0.75	6
12	0.917	1.000	1	6
10	1.150	1.250	1.25	6
8	1.375	1.500	1.5	6
6	1.333	1.500	2	6
5	1.300	1.500	2.5	6
4	1.750	2.000	3.5	6

- Find distance from back of the rack to the pinion centerline $B - C$

$$B - C = B + \frac{D_p}{2}$$

- Find the velocity of the rack V_{rack}

$$V_{\text{rack}} = \left(\frac{\pi}{6}\right) \left(\frac{D_p n_p}{2}\right)$$

- Find the time it takes to move the rack some distance

$$t = 60 \left(\frac{s_{\text{rack}}}{V_{\text{rack}}}\right)$$

- Find the number of revolutions required to move the rack that far

$$\theta_p = \left(\frac{6}{\pi}\right) \left(\frac{2s_{\text{rack}}}{D_p}\right)$$

2.6 Gear Trains

$$\text{train value: } TV = (VR)_1 (VR)_2 \cdots = \frac{N_{\text{output}}}{N_{\text{input}}} = \frac{n_{\text{input}}}{n_{\text{output}}}$$

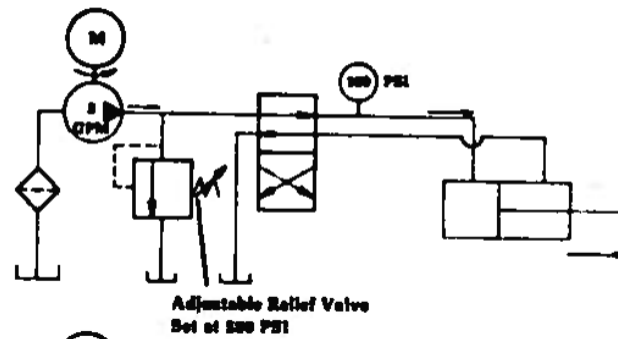
3 Fucking Fluids

3.1 Regenerative vs Non-Regenerative Circuits

There may be a question about the suitability of regenerative vs non regenerative circuits. They have their own advantages and disadvantages.

Summary: A regen circuit allows for more control over the ratio of force and speed for push and pull cycles. A non-regen circuit will always have a weaker/faster retraction cycle, which is ideal for application where pushing is the desired goal (hydraulic press). However if greater force is required for retraction, the regen circuit allows for balancing of push/pull forces, at the expense of greater complexity. see Section 3.2.3 for calculations.

Details: We examine a basic non-regenerative circuit first:



Usually the pump is at a fixed GPM flow rate and the system has a relief valve which regulates pressure to a constant value. The cycle above is in a state of pushing the piston and the fluid on the other side of the piston head provides no resistance to the pushing, so the full force of pressure times bore area is being applied. However when the piston retracts, it moves faster due to the rod occupying some of the bore volume on that side, with the same flow rate entering. So the result is a faster retraction at a lower force than the push cycle. No matter what bore and rod are selected there will be an imbalance in the cycles.

The regenerative circuit in contrast is able to produce equalized cycles:

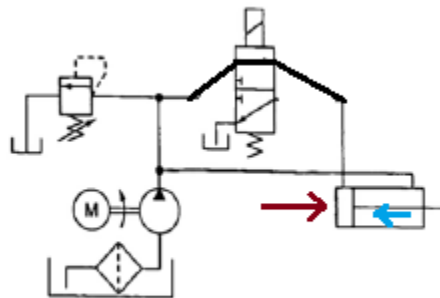


Figure 13-22 Regenerative circuit.

The push cycle is met by equal pressure on the other side. The force developed on the left in red is higher because the pressure area is larger, while the blue side is lower due to the rod. The net force developed is the pressure times the rod cross section area. The fluid will also recycle *regeneratively* through the system on the push cycle, allowing for more movement at a given volume of pump flow. On the pull cycle it operates the same as a non-regen circuit. The net effect is that the

ratio between the rod area and the (bore - rod) areas determines the ratio of movement velocity and force developed. If the ratio is one, then both the push and retraction speed/force are the same.

3.2 Selecting Bores and Pumps and Shit

3.2.1 Nomenclature

Nomenclature is not very well defined so here is what I will use:

s = distance (in)

v = speed (ft/min)

F = (total) force (lb)

F_w = force of weight (lb)

F_a = force from acceleration (lb)

F_f = force due to friction (lb)

μ = coefficient of friction

g = acceleration constant (from table. Not the same g as gravity)

p = pressure (psi)

P = power (hp)

L = stroke length (in)

L_b = basic length (in)

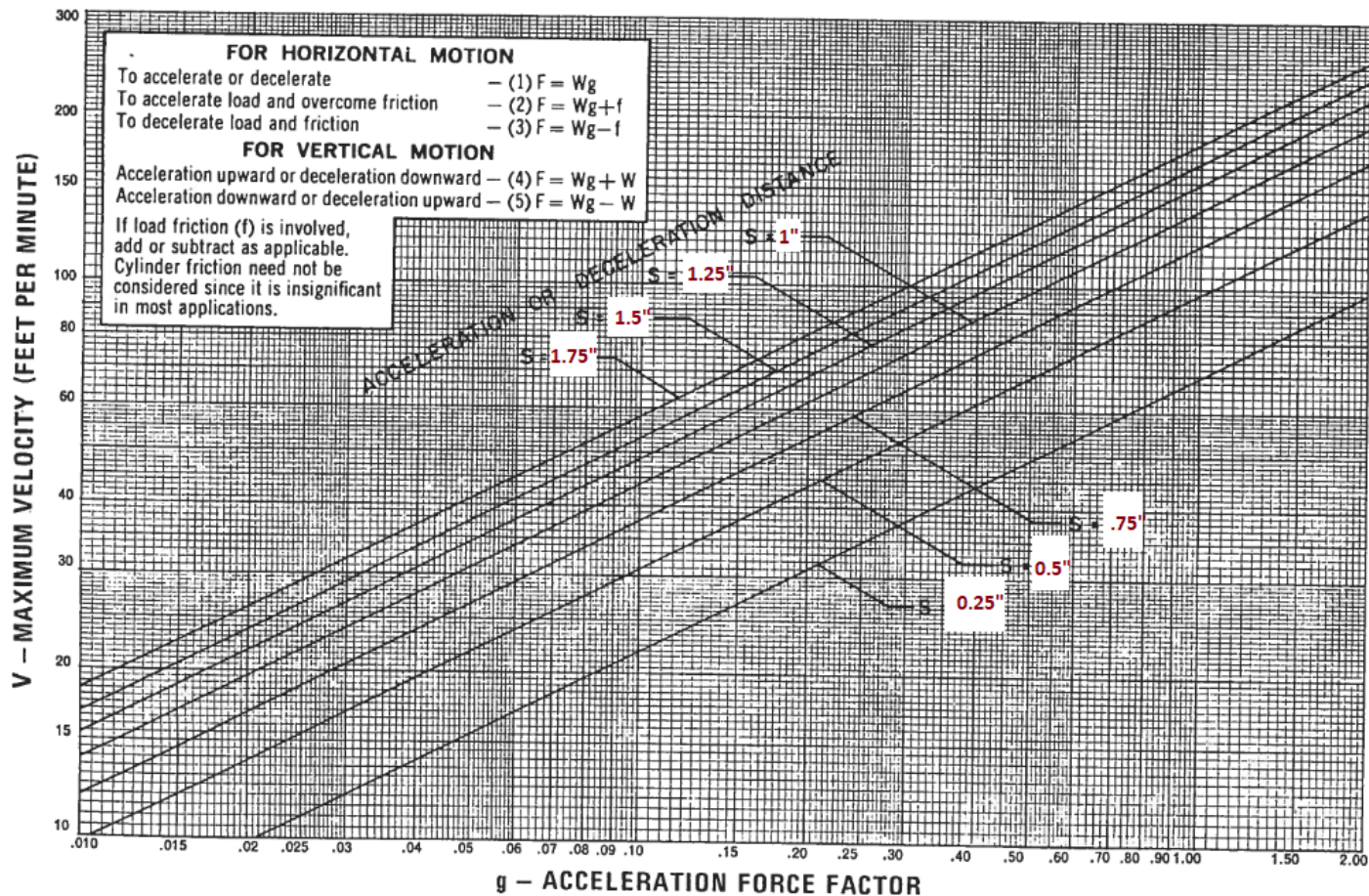
k = stroke factor

Q = flow rate (gpm)

3.2.2 Minimum Rod Diameter

1. Find the force acting on the piston (rod? The thingy that goes back and forward).
 - (a) Do some fucking free body analysis to find the relevant forces and velocities if you have a somewhat complicated system
 - (b) Find the force due to the weight of the load
 - (c) Find the force due to friction (if coefficient of friction is given, otherwise ignore that shit) Remember that the force will need to overcome friction to move so it is additive.
 - (d) If you are given that the rod has to approach some velocity, v , in some distance, s then we use the worst shitty photocopy table you've ever seen, using s (diagonals) and v (y-axis) to get g (x-axis).
 - (e) Compute g using the following equation or the table:

$$g = \left(\frac{v^2}{s} \right) (0.0000517) \text{ where } v \text{ is in ft/min and } s \text{ is in inches}$$



(f) Use g to compute the force due to the acceleration:

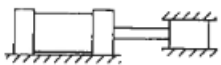
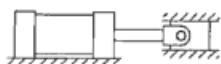
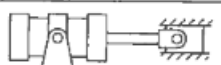
$$F_a = F_w g$$

(g) Sum the forces to get the total force (or thrust)

$$F = F_w + F_f + F_a$$

- The question should tell/show you how your pump is mounted. Use that to determine the stroke factor, k from this table

piston rod — stroke selection table

RECOMMENDED MOUNTING STYLES FOR MAXIMUM STROKE AND THRUST LOADS	ROD END CONNECTION	CASE	STROKE FACTOR
CLASS 1 — GROUPS 1 OR 3 Long stroke cylinders for thrust loads should be mounted using a heavy-duty mounting style at one end, firmly fixed and aligned to take the principle force. Additional mounting should be specified at the opposite end, which should be used for alignment and support. An intermediate support may also be desirable for long stroke cylinders mounted horizontally. Machine mounting pads can be adjustable for support mountings to achieve proper alignment.	FIXED AND RIGIDLY GUIDED.	I 	.50
	PIVOTED AND RIGIDLY GUIDED	II 	.70
	SUPPORTED BUT NOT RIGIDLY GUIDED	III 	2.00
CLASS 2 — GROUP 2 Style — Trunnion on Head	PIVOTED AND RIGIDLY GUIDED	IV 	1.00
Style — Intermediate Trunnion	PIVOTED AND RIGIDLY GUIDED	V 	1.50
Style — Trunnion on Cap or Style — Clevis on Cap	PIVOTED AND RIGIDLY GUIDED	VI 	2.00

3. Compute the basic length, L_b , from the stroke length, L (specified in question) and the stroke factor, k

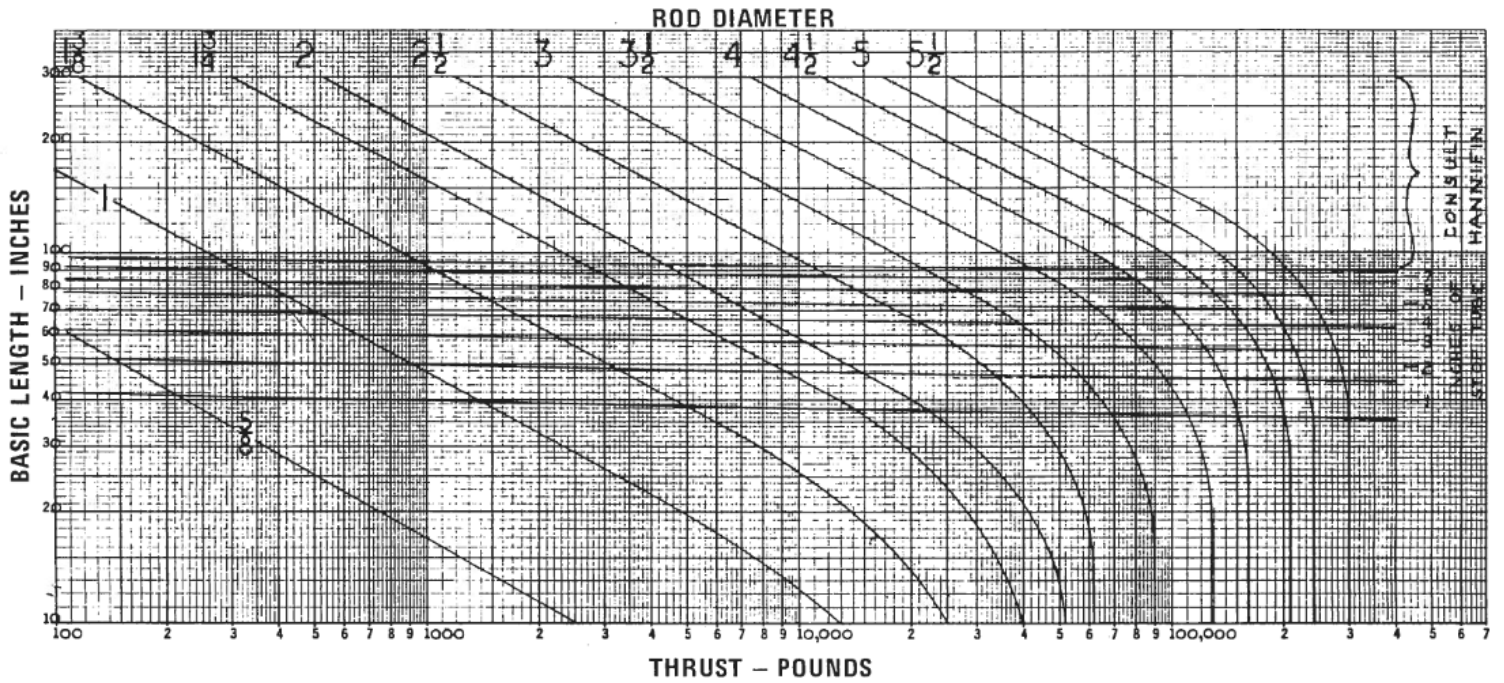
$$L_b = kL$$

4. Use the basic length and the total force (thrust) to choose the rod diameter. Find your point and round up to the nearest size. Because you may not be able to read this fucking awful table, the sizes are as follows:

$\frac{5}{8}$ ", 1", $1\frac{3}{8}$ ", $1\frac{3}{4}$ ", 2", $2\frac{1}{2}$ ", 3", $3\frac{1}{2}$ ", 4", $4\frac{1}{2}$ ", 5", $5\frac{1}{2}$ "

Note that this will give a minimum diameter, but the design requirements when it comes to balancing push and retraction or using regenerative circuits may require a larger bore diameter.

PISTON ROD – STROKE SELECTION GRAPH



3.2.3 Bore Size, Rod Size, Working Pressure

5. Use the pressure and the total force to choose a bore size from this table or calculate using formulas below. (Note that for pull applications, you must subtract force from the rod diameter)

theoretical push and pull forces for cylinders

CUSTOMARY U.S. UNITS

Cyl. Bore or Piston Rod. Dia. (in.)	Cyl. Bore Size (ϕ mm)	Area (sq. in.)	CYLINDER PUSH STROKE FORCE IN POUNDS AT VARIOUS PRESSURES (PSI)										Displacement per inch of Stroke (gallons)
			50	80	100	500	750	1000	1500	2000	2500	3000	
5/8	15.9	.307	15	25	31	154	230	307	461	614	768	921	.0013
1	25.4	.785	39	65	79	392	588	785	1,177	1,570	1,962	2,355	.0034
1-3/8	34.9	1.490	75	119	149	745	1,118	1,490	2,235	2,980	3,725	4,470	.0065
1-1/2	38.1	1.767	88	142	177	885	1,325	1,770	2,651	3,540	4,425	5,310	.00765
1-3/4	44.5	2.410	121	193	241	1,205	1,808	2,410	3,615	4,820	6,025	7,230	.0104
2	50.8	3.140	157	251	314	1,570	2,357	3,140	4,713	6,280	7,850	9,420	.0136
2-1/2	63.5	4.910	245	393	491	2,455	3,682	4,910	7,364	9,820	12,275	14,730	.0213
3	76.2	7.070	354	566	707	3,535	5,302	7,070	10,604	14,140	17,675	21,210	.0306
3-1/4	82.6	8.300	415	664	830	4,150	6,225	8,300	12,450	16,600	20,750	24,900	.0359
3-1/2	88.9	9.620	481	770	962	4,810	7,215	9,620	14,430	19,240	24,050	28,860	.0416
4	101.6	12.570	628	1,006	1,257	6,285	9,428	12,570	18,856	25,140	31,425	37,710	.0544
5	127.0	19.640	982	1,571	1,964	9,820	14,730	19,640	29,460	39,280	49,100	58,920	.0850
5-1/2	139.7	23.760	1,188	1,901	2,376	11,880	17,820	23,760	35,640	47,520	59,400	71,280	.1028
6	152.4	28.270	1,414	2,262	2,827	14,135	21,203	28,270	42,406	56,540	70,675	84,810	.1224
7	177.8	38.490	1,924	3,079	3,849	19,245	28,868	38,490	57,736	76,980	96,225	115,470	.1666
8	203.2	50.270	2,513	4,022	5,027	25,135	37,703	50,270	75,406	100,540	125,675	150,810	.2176
8-1/2	215.9	56.750	2,838	4,540	5,675	28,375	42,563	56,750	85,125	113,500	142,875	170,250	.2455
10	254.0	78.540	3,927	6,283	7,854	39,270	58,905	78,540	117,810	157,080	196,350	235,620	.3400
12	304.8	113.100	5,655	9,048	11,310	56,550	84,825	113,100	169,650	226,200	282,750	339,300	.4896

table b-1

NOTE: Deduct Force of Piston Rod Size from Bore Size for Pull Applications.

SI (METRIC) UNITS

Cyl. Bore or Piston Rod Dia. (in.)	Size in MM	Area in Sq. MM	CYLINDER PUSH FORCE IN NEWTONS AT VARIOUS PRESSURES IN BARS										Displacement for 1 MM of Stroke (Cu. MM)
			4	6.3	10	16	25	40	63	100	160	200	
5/8	15.87	197.9	79	125	198	317	495	792	1247	1979	3167	3959	197.9
1	25.40	506.7	203	319	507	811	1267	2027	3192	5067	8107	10134	506.7
1-3/8	34.93	958.0	383	604	958	1533	2395	3832	6035	9580	15328	19160	958.0
1-1/2	38.10	1140.1	456	718	1140	1824	2850	4560	7183	11401	18242	22802	1140.1
1-3/4	44.45	1551.8	621	978	1552	2483	3879	6207	9776	15518	24829	31036	1551.8
2	50.80	2026.9	811	1277	2027	3243	5067	8107	12769	20268	32429	40537	2026.9
2-1/2	63.50	3166.9	1267	1995	3167	5067	7917	12668	19952	31669	50671	63339	3166.9
3	76.20	4560.4	1824	2873	4560	7297	11401	18242	28730	45604	72966	91208	4560.4
3-1/4	82.55	5352.1	2141	3372	5352	8563	13380	21408	33718	53521	85634	107042	5352.1
3-1/2	88.90	6207.2	2483	3911	6207	9931	15518	24829	39105	62072	99315	124144	6207.2
4	101.60	8107.3	3243	5108	8107	12972	20268	32429	51076	81073	129717	162147	8107.3
5	127.00	12667.7	5067	7981	12668	20268	31669	50671	79807	126677	202683	253354	12667.7
5-1/2	139.70	15327.9	6131	9657	15328	24525	38320	61312	96566	153279	245247	306559	15327.9
6	152.40	18241.5	7297	11492	18242	29186	45604	72966	114922	182415	291864	364830	18241.5
7	177.80	24828.1	9931	15642	24829	39726	62072	99315	156421	248287	397260	496574	24828.7
8	203.20	32429.4	12972	20430	32429	51887	81073	129717	204305	324294	518870	648587	32429.4
8-1/2	215.90	36609.7	14644	23064	36610	58576	91524	146439	230641	366097	585755	732194	36609.7
10	254.00	50670.9	20268	31923	50671	81073	126677	202683	319226	506709	810734	1013417	50670.9
12	304.80	72966.0	29186	45968	72966	116746	182415	291864	459686	729660	1167457	1459321	72966.0

table b-2

REF. 1 #f = 4.448 NEWTONS (N)
1 BAR = 14.504 PSI

Alternatively, you can apply the principle of $F = PA$ to get rod and bore diameter to force relations.

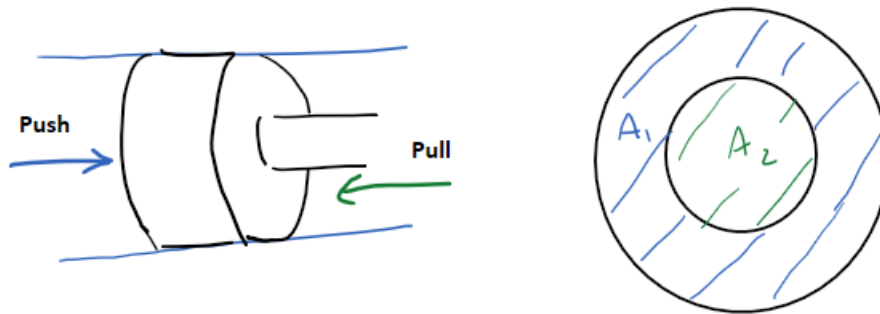


Figure 1: Piston Area and Motion

From the diagram above we can relate the areas shown to the forces developed, depending on if the circuit is *regenerative* or not. The advantage to a regenerative circuit is that both the push and pull forces can be made equal by making $A_1 = A_2$

Solve for any push pull force ratio constraints using standardized rod and bore sizes from the tables, making sure to keep a minimum rod size larger than the buckling requirement from

earlier. The relation between force in lbs and pressure in psi is given below, use it to size parts or determine a working pressure required.

$$A_1 = \frac{\pi}{4}(d_{bore}^2 - d_{rod}^2)$$

$$A_2 = \frac{\pi}{4}d_{rod}^2$$

	Regular	Regenerative
Push	$F = P \cdot (A_1 + A_2)$	$F = P \cdot (A_2)$
Pull	$F = P \cdot (A_1)$	$F = P \cdot (A_1)$

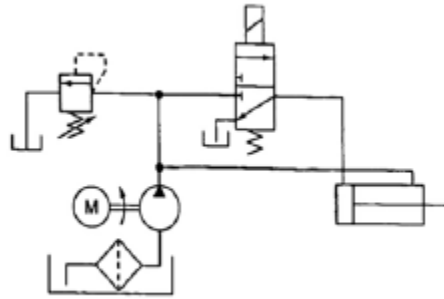


Figure 13-22 Regenerative circuit.

If using a regular push piston with known pressure, the required bore diameter can be found:

$$d_{bore} = \sqrt{\frac{4F}{\pi P}}$$

and then round the value up to the nearest standard size.

You can then compute the rated force using the diameter you chose;

$$F = \frac{\pi}{4}P d_{bore}^2$$

3.2.4 Volume Flow Rate

6. Convert v to ft/min if not already
7. Compute the pump capacity, Q in gpm using v in ft/min and A in in².

$$Q_p = v_p A_p \left(\frac{12}{231} \right) = \frac{\pi}{4} v_p d_p^2 \left(\frac{12}{231} \right)$$

	Regular	Regenerative
Push	$Q = \frac{\pi}{4} v_{push} d_{bore}^2 \left(\frac{12}{231} \right)$	$Q = \frac{\pi}{4} v_{pull} (d_{rod}^2) \left(\frac{12}{231} \right)$
Pull	$Q = \frac{\pi}{4} v_{push} (d_{bore}^2 - d_{rod}^2) \left(\frac{12}{231} \right)$	$Q = \frac{\pi}{4} v_{push} (d_{bore}^2 - d_{rod}^2) \left(\frac{12}{231} \right)$

3.2.5 Pipe Sizing

8. Compute Pipe Diameter

There are two methods to select pipe sizing. We always use a maximum pipeflow velocity of 15 ft/s so we should find minimum pipe size for this constraint:

$$d_p = \sqrt{\frac{4Q}{\pi v} \left(\frac{231}{720} \right)}$$

where $v = 15\text{ft/s}$ and Q is in GPM. Round this value up to the nearest standard size from the table to select a pipe size.

9. Method 1: Using the table from Fluid Power Basics: The working pressure determines the schedule, 40 - Low, 80 - Medium, 160 - High. Just assume Schedule 80 saying that proper schedule is dependent on working pressure and material selection.

Get the pipe diameter in the blue box under Schedule 80 that is larger than the minimum calculated in the last step. The nominal pipe size and Schedule can be specified for the pipe.

HYDRAULIC PIPE SPECIFICATIONS (Dimensions)				
Nominal Size	Pipe OD	INSIDE DIAMETER		
		Schedule 40	Schedule 80	Schedule 160
$\frac{1}{8}$	.405	.289	.215	
$\frac{1}{4}$	.540	.384	.302	
$\frac{3}{8}$	.675	.493	.423	
$\frac{1}{2}$	.840	.622	.554	.466
$\frac{3}{4}$	1.050	.824	.742	.587
1	1.315	1.049	.957	.815
1 $\frac{1}{4}$	1.660	1.380	1.278	1.160
1 $\frac{1}{2}$	1.900	1.610	1.500	1.338
2	2.375	2.067	1.939	1.859
2 $\frac{1}{2}$	2.875	2.469	2.323	2.125
3	3.500	3.068	2.900	2.624

10. Method 2: Alternatively try selecting a pipe size from this table:

CYLINDER BORE-INCHES	PISTON ROD		CYLINDER NET AREA SQ. IN.	FLUID DISPLACEMENT AT 10 FT. PER MINUTE PISTON VELOCITY		FLUID VELOCITY (IN FEET PER SECOND) THROUGH EXTRA HEAVY PIPE AT 10 F.P.M. PISTON SPEED.									
	DIA.-INCHES	AREA SQ. IN.				G.P.M.	C.F.M.	1/4	3/8	1/2	3/4	1	1-1/4	1-1/2	2
1	0	0	0.785	0.41	0.054	1.82	0.92	0.56	0.30	0.183	0.102	0.074	0.045		
	1/2	0.196	0.589	0.30	0.041	1.33	0.68	0.41	0.21	0.134	0.075	0.055	0.033		
	5/8	0.307	0.478	0.16	0.033	0.71	0.36	0.22	0.12	0.071	0.040	0.029	0.017		
1½	0	0	1.77	0.92	0.123	4.09	2.09	1.259	0.680	0.410	0.230	0.167	0.100		
	5/8	0.307	1.46	0.76	0.101	3.38	1.73	1.040	0.562	0.338	0.190	0.138	0.082		
	1	0.785	0.98	0.51	0.068	2.27	1.16	0.699	0.378	0.228	0.128	0.093	0.055		
2	0	0	3.14	1.63	0.218	7.27	3.71	2.238	1.209	0.728	0.408	0.296	0.177		
	5/8	0.307	2.84	1.48	0.197	6.56	3.35	2.019	1.091	0.657	0.368	0.267	0.160		
	1	0.785	2.36	1.23	0.164	5.45	2.79	1.678	0.907	0.546	0.306	0.222	0.133		
2½	1-3/8	1.485	1.66	0.86	0.115	3.84	1.96	1.180	0.638	0.384	0.215	0.156	0.094		
	0	0	4.91	2.55	0.341	11.36	5.80	3.496	1.890	1.138	0.638	0.463	0.277		
	5/8	0.307	4.60	2.39	0.319	10.65	5.44	3.278	1.771	1.067	0.598	0.434	0.260		
3	1	0.785	4.12	2.14	0.286	9.54	4.87	2.937	1.587	0.956	0.536	0.389	0.233		
	1-3/8	1.485	3.42	1.78	0.237	7.93	4.05	2.439	1.318	0.794	0.445	0.323	0.193		
	1-3/4	2.405	2.50	1.30	0.174	5.96	2.96	1.783	0.963	0.580	0.325	0.236	0.141		
3½	0	0	8.30	4.31	0.576	19.20	9.81	5.909	3.193	1.923	1.078	0.783	0.468		
	1	0.785	7.51	3.90	0.521	17.38	8.88	5.349	2.891	1.741	0.976	0.708	0.424		
	1-3/8	1.485	6.81	3.54	0.473	15.77	8.05	4.851	2.622	1.579	0.885	0.642	0.384		
4	1-3/4	2.405	5.89	3.06	0.409	13.64	6.96	4.196	2.268	1.366	0.765	0.556	0.333		
	2	3.142	5.15	2.68	0.357	11.93	6.09	3.671	1.984	1.195	0.670	0.486	0.291		
	0	0	12.57	6.53	0.872	29.09	14.85	8.95	4.84	2.91	1.63	1.19	0.709		
5	1	0.785	11.78	6.12	0.818	27.27	13.93	8.39	4.54	2.73	1.53	1.11	0.665		
	1-3/8	1.485	11.08	5.76	0.769	25.65	13.10	7.89	4.27	2.57	1.44	1.05	0.625		
	1-3/4	2.405	10.16	5.28	0.705	23.52	12.01	7.24	3.91	2.36	1.32	0.96	0.574		
6	2	3.142	9.42	4.89	0.654	21.82	11.14	6.71	3.63	2.19	1.22	0.89	0.532		
	2-1/2	4.909	7.66	3.98	0.532	17.73	9.05	5.45	2.95	1.78	1.00	0.72	0.432		
	0	0	19.64	10.20	1.363	45.45	23.21	13.99	7.56	4.55	2.55	1.85	1.108		
7	1	0.785	18.85	9.79	1.308	43.64	22.28	13.43	7.26	4.37	2.45	1.78	1.064		
	1-3/8	1.485	18.15	9.43	1.260	42.01	21.45	12.93	6.99	4.21	2.36	1.71	1.024		
	1-3/4	2.405	17.23	8.95	1.196	39.88	20.37	12.27	6.63	3.99	2.24	1.63	0.973		
8	2	3.142	16.49	8.57	1.144	38.18	19.50	11.75	6.35	3.82	2.14	1.56	0.931		
	2-1/2	4.909	14.73	7.65	1.022	34.09	17.41	10.49	5.67	3.41	1.91	1.39	0.831		
	3	7.069	12.57	6.53	0.872	29.09	14.85	8.95	4.84	2.91	1.63	1.19	0.709		
9	3-1/2	9.621	10.01	5.21	0.695	23.18	11.84	7.13	3.86	2.32	1.30	0.95	0.565		
	0	0	28.27	14.69	1.962	65.45	33.42	20.14	10.88	6.55	3.67	2.67	1.596		
	1-3/8	1.485	26.79	13.92	1.859	62.01	31.67	19.08	10.31	6.21	3.48	2.53	1.512		
10	1-3/4	2.405	25.87	13.44	1.795	59.88	30.58	18.43	9.96	5.60	3.36	2.44	1.460		
	2	3.142	25.13	13.06	1.744	58.18	29.71	17.90	9.67	5.83	3.27	2.37	1.418		
	2-1/2	4.909	23.37	12.14	1.622	54.1	27.6	16.64	8.99	5.42	3.04	2.20	1.32		
11	3	7.069	21.21	11.02	1.472	49.1	25.1	15.10	8.16	4.92	2.76	2.00	1.20		
	3-1/2	9.621	18.65	9.69	1.294	43.2	22.1	13.29	7.18	4.32	2.42	1.76	1.05		
	4	12.566	15.71	8.16	1.090	36.4	18.6	11.19	6.05	3.64	2.04	1.48	0.89		

7	0	0	38.49	20.00	2.671	89.1	45.5	27.41	14.81	8.92	5.00	3.63	2.17	
	1-3/8	1.485	37.00	19.22	2.568	85.7	43.7	26.35	14.24	8.58	4.81	3.49	2.09	
	1-3/4	2.405	36.08	18.74	2.504	83.5	42.7	25.70	13.89	8.36	4.69	3.40	2.04	
	2	3.142	35.34	18.36	2.453	81.8	41.8	25.17	13.60	8.19	4.59	3.33	2.00	
	2-1/2	4.909	33.58	17.44	2.330	77.7	39.7	23.92	12.92	7.78	4.36	3.17	1.90	
	3	7.069	31.42	16.32	2.181	72.7	37.1	22.38	12.09	7.28	4.08	2.96	1.77	
	3-1/2	9.621	28.86	14.99	2.003	66.8	34.1	20.56	11.11	6.69	3.75	2.72	1.63	
	4	12.566	25.92	13.47	1.799	60.0	30.6	18.46	9.98	6.01	3.37	2.45	1.46	
	4-1/2	15.904	22.58	11.73	1.567	52.3	26.7	16.08	8.69	5.23	2.93	2.12	1.28	
	5	19.635	18.85	9.79	1.308	43.6	22.3	13.43	7.26	4.37	2.45	1.78	1.06	
8	0	0	50.27	26.12	3.489	116.4	59.4	35.80	19.35	11.65	6.53	4.74	2.84	1.977
	1-3/8	1.485	48.78	25.34	3.385	112.9	57.7	34.74	18.78	11.31	6.34	4.60	2.75	1.918
	1-3/4	2.405	47.86	24.86	3.321	110.8	56.6	34.09	18.42	11.09	6.22	4.51	2.70	1.882
	2	3.142	47.12	24.48	3.270	109.1	55.7	33.56	18.14	10.92	6.12	4.45	2.66	1.853
	2-1/2	4.909	45.36	23.57	3.149	105.0	53.61	32.31	17.46	10.51	5.892	4.278	2.560	1.784
	3	7.069	43.20	22.44	2.998	100.0	51.06	30.77	16.63	10.01	5.612	4.074	2.438	1.699
	3-1/2	9.621	40.65	21.12	2.821	94.1	48.04	28.95	15.65	9.42	5.279	3.834	2.294	1.598
	4	12.566	37.70	19.59	2.616	87.3	44.56	26.85	14.51	8.74	4.897	3.556	2.128	1.483
	4-1/2	15.904	34.36	17.85	2.385	79.5	40.62	24.47	13.23	8.20	4.464	3.241	1.939	1.351
	5	19.635	30.63	15.91	2.126	70.9	36.21	21.82	11.79	7.10	3.979	2.889	1.729	1.205
10	5-1/2	23.758	26.51	13.77	1.840	61.4	31.33	18.88	10.20	6.15	3.444	2.500	1.496	1.043
	0	0	78.54	40.80	5.451	181.8	92.84	55.94	30.23	18.21	10.203	7.408	4.433	3.089
	1-3/4	2.405	76.14	39.56	5.284	176.2	89.99	54.23	29.31	17.65	9.890	7.181	4.297	2.994
	2	3.142	75.40	39.17	5.233	174.5	89.12	53.70	29.02	17.48	9.795	7.112	4.255	2.965
	2-1/2	4.909	73.63	38.25	5.110	170.4	87.03	52.44	28.34	17.07	9.565	6.945	4.156	2.896
	3	7.069	71.47	37.13	4.960	165.4	84.48	50.91	27.51	16.57	9.284	6.741	4.034	2.811
	3-1/2	9.621	68.92	35.80	4.783	159.5	81.47	49.09	26.53	15.98	8.953	6.501	3.890	2.710
	4	12.566	65.97	34.27	4.578	152.7	77.98	46.99	25.39	15.29	8.570	6.223	3.724	2.595
	4-1/2	15.904	62.64	32.54	4.347	145.0	74.04	44.61	24.11	14.52	8.137	5.908	3.535	2.463
	5	19.635	58.91	30.60	4.088	136.4	69.63	41.96	22.67	13.65	7.652	5.556	3.325	2.317
12	5-1/2	23.758	54.78	28.46	3.802	126.8	64.75	39.02	21.09	12.70	7.116	5.167	3.092	2.154
	6	28.274	50.27	26.12	3.489	116.4	59.42	35.80	19.35	11.65	6.530	4.741	2.837	1.977
	6-1/2	33.183	45.36	23.57	3.148	105.0	53.6	32.31	17.46	10.52	5.89	4.278	2.560	1.784
	7	38.485	40.06	20.81	2.780	92.7	47.4	28.53	15.42	9.29	5.20	3.778	2.261	1.575
	0	0	113.10	58.76	7.849	261.8	133.7	80.55	43.53	26.22	14.69	10.668	6.383	4.448
	2	3.142	109.96	57.12	7.631	254.5	130.0	78.32	42.32	25.49	14.28	10.371	6.206	4.324
	2-1/2	4.909	108.19	56.21	7.508	250.4	127.9	77.06	41.64	25.08	14.05	10.205	6.106	4.255
	3	7.069	106.03	55.08	7.359	245.4	125.3	75.52	40.81	24.58	13.77	10.001	5.984	4.170
	3-1/2	9.621	103.48	53.76	7.182	239.5	122.3	73.70	39.83	23.99	13.44	9.760	5.840	4.069
	4	12.566	100.53	52.23	6.977	232.7	118.8	71.60	38.70	23.30	13.06	9.482	5.674	3.954
14	4-1/2	15.904	97.19	50.49	6.745	225.0	114.9	69.23	37.41	22.53	12.63	9.168	5.486	3.822
	5	19.635	93.46	48.55	6.486	216.4	110.5	66.57	35.98	21.67	12.14	8.816	5.275	3.676
	5-1/2	23.758	89.34	46.41	6.200	206.8	105.6	63.63	34.39	20.71	11.61	8.427	5.042	3.513
	6	28.274	84.82	44.06	5.887	196.4	100.3	60.42	32.65	19.66	11.02	8.001	4.787	3.336
	6-1/2	33.183	79.92	41.52	5.547	185.0	94.5	56.92	30.76	18.53	10.38	7.538	4.510	3.143
	7	38.485	74.61	38.77	5.179	172.7	88.2	53.14	28.72	17.30	9.69	7.038	4.211	2.934
	7-1/2	44.179	68.92	35.80	4.783	159.5	81.5	49.09	26.53	15.98	8.95	6.501	3.890	2.710
	8	50.266	62.83	32.64	4.360	145.4	74.3	44.75	24.19	14.57	8.16	5.926	3.546	2.471
	8-1/2	56.745	56.35	29.27	3.911	130.5	66.6	40.14	21.69	13.06	7.32	5.315	3.181	2.216
	0	0	153.94	79.97	10.683	356.3	182.0	109.6	59.25	35.68	20.00	14.52	8.688	6.054
14	2-1/2	4.909	149.03	77.42	10.343	345.0	176.2	106.2	57.36	34.55	19.36	14.06	8.411	5.861
	3	7.069	146.87	76.30	10.193	340.0	173.6	104.6	56.53	34.05	19.08	13.85	8.289	5.776
	3-1/2	9.621	144.32	74.97	10.016	334.1	170.6	102.8	55.55	33.45	18.75	13.61	8.145	5.676
	4	12.566	141.37	73.44	9.811	327.3	167.1	100.7	54.42	32.77	18.37	13.33	7.979	5.560
	4-1/2	15.904	138.03	71.71	9.579	319.5	163.2	98.3	53.13	32.00	17.93	13.02	7.791	5.428
	5	19.635	134.30	69.77	9.320	310.9	158.8	95.7	51.70	31.13	17.45	12.67	7.580	5.282
	5-1/2	23.758	130.18	67.63	9.035	301.3	153.9	92.7	50.11	30.18	16.91	12.28	7.347	5.120

3.2.6 Horsepower for Motor

11. Compute the required horsepower. Or use a fucking table. Same difference

$$HP = \frac{Q \cdot P}{1714 \cdot 0.85}$$

Note that Q is the flow rate of the stroke, P is the pressure, 0.85 is the assumed efficiency, and 1714 is a conversion factor between GPM and HP.

ELECTRIC MOTOR HORSEPOWER REQUIRED TO DRIVE A HYDRAULIC PUMP

GPM	100 PSI	200 PSI	250 PSI	300 PSI	400 PSI	500 PSI	750 PSI	1000 PSI	1250 PSI	1500 PSI	2000 PSI	2500 PSI	3000 PSI
1/2	.04	.07	.09	.11	.14	.18	.26	.35	.44	.53	.70	.88	1.10
1	.07	.14	.18	.21	.28	.35	.52	.70	.88	1.05	1.40	1.76	1.92
1-1/2	.10	.21	.26	.31	.41	.52	.77	1.03	1.29	1.55	2.06	2.58	3.09
2	.14	.28	.35	.42	.56	.70	1.04	1.40	1.76	2.10	2.80	3.53	4.20
2-1/2	.17	.34	.43	.51	.69	.86	1.29	1.72	2.15	2.58	3.44	4.30	5.14
3	.21	.42	.53	.63	.84	1.05	1.56	2.10	2.64	3.15	4.20	5.28	6.30
3-1/2	.24	.48	.60	.72	.96	1.20	1.80	2.40	3.00	3.60	4.80	6.00	7.20
4	.28	.56	.70	.84	1.12	1.40	2.08	2.80	3.52	4.20	5.60	7.04	8.40
5	.35	.70	.88	1.05	1.40	1.75	2.60	3.50	4.40	5.25	7.00	8.80	10.50
6	.42	.84	1.05	1.26	1.68	2.10	3.12	4.20	5.28	6.30	8.40	10.56	12.60
7	.49	.98	1.23	1.47	1.96	2.45	3.64	4.90	6.16	7.35	9.80	12.32	14.70
8	.56	1.12	1.40	1.68	2.24	2.80	4.16	5.60	7.04	8.40	11.20	14.08	16.80
9	.62	1.24	1.55	1.86	2.48	3.10	4.65	6.18	7.73	9.28	12.40	15.56	18.58
10	.70	1.40	1.75	2.10	2.80	3.50	5.20	7.00	8.80	10.50	14.00	17.60	21.00
11	.77	1.54	1.93	2.31	3.08	3.85	5.72	7.70	9.68	11.50	15.40	19.36	23.10
12	.84	1.68	2.10	2.52	3.36	4.20	6.24	8.40	10.50	12.60	16.80	21.00	25.20
13	.89	1.78	2.23	2.67	3.56	4.45	6.68	8.92	11.20	13.40	17.80	22.40	26.72
14	.96	1.92	2.40	2.88	3.84	4.80	7.20	9.60	12.00	14.40	19.20	24.00	28.80
15	1.05	2.10	2.63	3.15	4.20	5.25	7.80	10.50	13.20	15.70	21.00	26.40	31.50
16	1.10	2.20	2.75	3.30	4.40	5.50	8.25	11.00	13.80	16.50	22.00	27.60	33.00
17	1.17	2.34	2.93	3.51	4.68	5.85	8.78	11.70	14.60	17.60	23.40	29.20	35.10
18	1.26	2.52	3.15	3.78	5.04	6.30	9.35	12.60	15.80	18.90	25.20	31.60	37.80
19	1.30	2.60	3.25	3.90	5.20	6.50	9.75	13.00	16.30	19.50	26.00	32.60	39.00
20	1.40	2.80	3.50	4.20	5.60	7.00	10.40	14.00	17.60	21.00	28.00	35.20	42.00
25	1.75	3.50	4.38	5.25	7.00	8.75	13.10	17.50	21.90	26.20	35.00	43.80	52.50
30	2.10	4.20	5.25	6.30	8.40	10.50	15.60	21.00	26.40	31.50	42.00	52.80	63.00
35	2.45	4.90	6.13	7.35	9.80	12.20	18.40	24.50	30.60	36.70	49.00	61.20	73.50
40	2.80	5.60	7.00	8.40	11.20	14.00	20.80	28.00	35.20	42.00	56.00	70.40	84.00
45	3.15	6.30	7.87	9.45	12.60	15.80	23.60	31.50	39.40	47.30	63.00	78.80	94.50
50	3.50	7.00	8.75	10.50	14.00	17.50	26.00	35.00	44.00	52.50	70.00	88.00	105.00
55	3.85	7.70	9.63	11.60	15.40	19.30	28.60	38.50	48.40	57.80	77.00	96.80	115.50
60	4.20	8.40	10.50	12.60	16.80	21.00	31.20	42.00	52.80	63.00	84.00	105.60	126.00
65	4.55	9.10	11.40	13.60	18.20	22.80	33.80	45.50	57.20	68.20	90.00	114.40	136.50

12. Based on your required horsepower, choose a motor to use. Three phase power is for industrial applications and would be available in that setting.

3 PHASE MOTOR STARTERS																						
1/2 TO 20 H.P. MOTOR H.P. 3 ∅	1/2		3/4		1		1-1/2		2		3		5		7-1/2		10		15		20	
	VOLTAGE																					
	220	440	220	440	220	440	220	440	220	440	220	440	220	440	220	440	220	440	220	440	220	440
Nema Starter Size	00	00	00	00	00	00	00	00	0	00	0	0	1	0	1	1	2	1	2	2	3	2
⊗ Full Load Current	2.0	1.0	2.8	1.4	3.5	1.8	5.0	2.5	6.5	3.3	9.0	4.5	15	7.5	22	11	28	14	40	20	52	26
Fuses — Amps { Std. N.E.C.	15	15	15	15	15	15	15	15	20	15	25	15	40	20	60	30	70	35	100	50	150	70
Dual Element	5	4	5	4	8	4	8	5	12	8	15	10	25	15	30	20	45	25	60	30	80	40
Circuit Breaker Max. Amps.	15	15	15	15	15	15	15	15	15	15	20	15	30	15	50	20	50	30	70	40	100	50
Minimum Wire Sizes { R, RW, T, TW	14	14	14	14	14	14	14	14	14	14	14	14	12	14	10	14	8	12	6	10	4	8
RH	14	14	14	14	14	14	14	14	14	14	14	14	12	14	10	14	8	12	6	10	6	8
Always specify voltage and frequency.																						

3 PHASE MOTOR STARTERS																				
25 TO 200 H.P. MOTOR H.P. 3 ϕ	25		30		40		50		60		75		100		125		150		200	
	220	440	220	440	220	440	220	440	220	440	220	440	220	440	220	440	220	440	220	440
Nema Starter Size	3	2	3	3	4	3	4	3	5	4	5	4	5	4	6	5	6	5	6	5
⊗ Full Load Current	64	32	78	39	104	52	125	63	150	75	185	93	246	123	310	155	360	180	480	240
Fuses — Amps { Std. N.E.C.	175	80	200	100	300	150	350	175	400	200	500	250	600	350	—	400	—	450	—	600
Dual Element	100	50	125	60	175	80	200	100	225	125	300	150	400	200	450	250	600	300	—	400
Circuit Breaker Max. Amps.	125	50	100	70	175	100	200	125	225	125	300	150	400	200	—	250	—	300	—	400
Minimum Wire Sizes { R, RW, T, TW	3	8	1	6	00	4	000	3	0000	2	300	0	500	000	—	0000	—	300	—	500
	4	8	3	6	1	6	00	4	000	3	0000	1	350	00	—	000	—	0000	—	350
Always specify voltage and frequency.																				

SINGLE PHASE MOTOR STARTERS																						
1/6 TO 5 H.P.	MOTOR H.P. 1 ∅		1/6		1/4		1/3		1/2		3/4		1		1-1/2		2		3		5	
	VOLTAGE		115	230	115	230	115	230	115	230	115	230	115	230	115	230	115	230	115	230	115	230
⊗ Full Load Current		4.4	2.2	5.8	2.9	7.2	3.6	9.8	4.9	13.8	6.9	16	8	20	10	24	12	34	17	56	28	
Fuses – Amps. Std. N.E.C.		15	15	20	15	25	15	30	15	45	25	50	25	60	30	80	40	100	60	—	90	
Circuit Breaker Max. Amps.		15	15	15	15	15	15	30	15	40	20	40	20	50	30	70	30	100	50	—	70	
Min. Wire Sizes – R, RH, RW, T, TW		14	14	14	14	14	14	14	14	12	14	12	14	10	14	10	14	6	10	—	8	

WIRE & CONDUIT SIZES																						
WIRE SIZE AWG or MCM		14	12	10	8	6	4	3	2	1	0	00	000	0000	250	300	350	400	500	750	1000	
MAXIMUM 3 WIRES IN CONDUIT	Wire Capacity { R-RW-T-TW Amps. RH Amps.	15	20	30	40	55	70	80	95	110	125	145	165	195	215	240	260	280	320	400	455	
		15	20	30	45	65	85	100	115	130	150	175	200	230	255	285	310	335	380	475	545	
	CONDUIT SIZE – Inches	1/2	1/2	3/4	3/4	1	1-1/4	1-1/4	1-1/4	1-1/2	2	2	2	2-1/2	2-1/2	2-1/2	3	3	3	3-1/2	4	
Volts Drop Per Ampere { 1 Phase Volts		.4762	.3125	.1961	.1250	.0833	.0538	.0431	.0370	.0323	.0269	.0222	.0190	.0161	.0147	.0131	.0121	.0115	.0101	.0086	.0081	
Per 100 Ft. – 80% P.F. 3 Phase Volts		.4167	.2632	.1677	.1087	.0714	.0463	.0379	.0323	.0278	.0231	.0196	.0163	.0139	.0128	.0114	.0106	.0091	.0088	.0066	.0061	

Capacity of conductors in conduit based on room temperature of 30° C. (86° F.)

⊗ The full load currents shown are average values.

3.3 Circuit Analysis Techniques

3.3.1 Flowrate

The following are some handy fluid relations to use when computing flow rates and power:

$$Q = Av$$

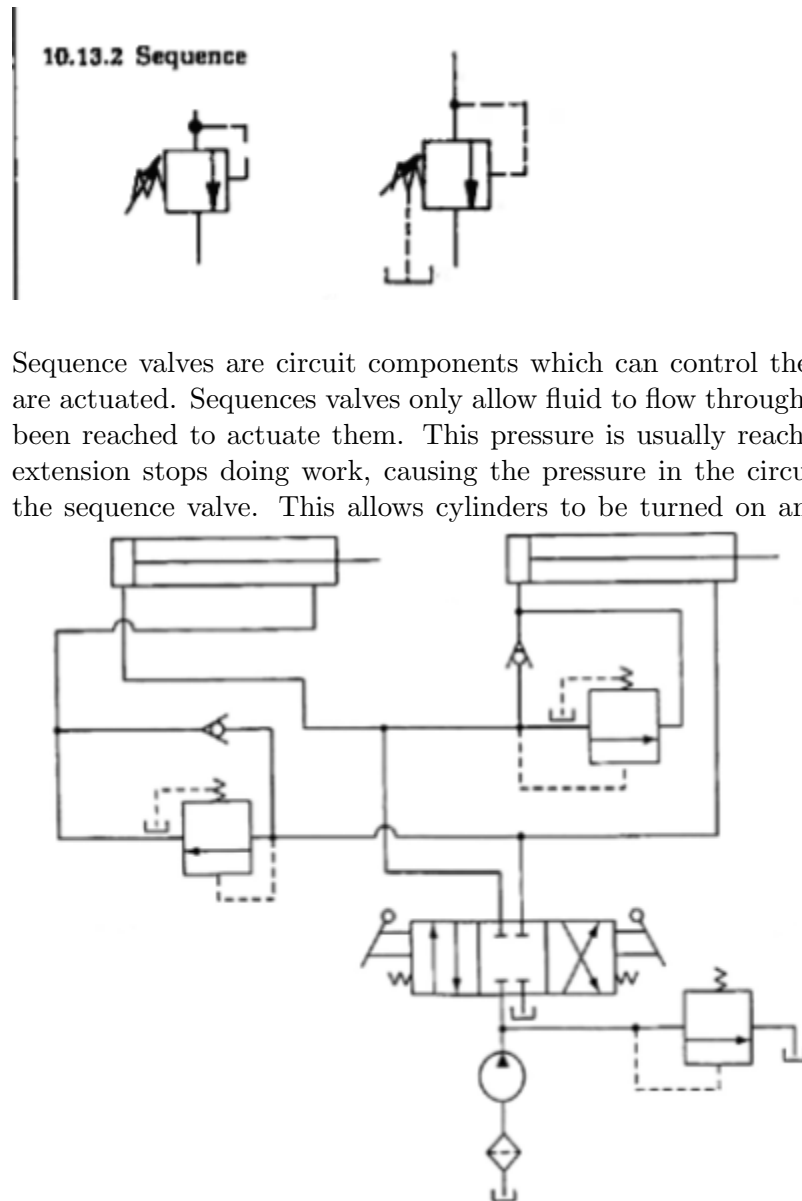
Use the area of the fluid column and the movement speed of the piston head to get Q (usually in in^3) then convert to GPM with $\frac{in^3}{s} = \frac{60}{231} \text{ GPM}$

3.3.2 Power

Work is pressure times volume, so power is the time derivative. Conversion factor for Q in GPM and P in psi included below.

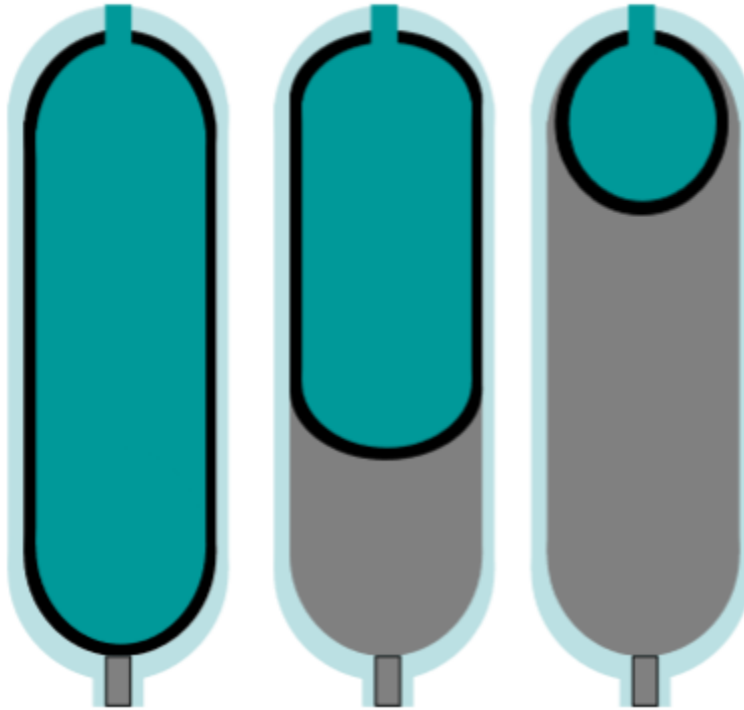
$$\text{Power} = P \frac{dV}{dt} = \frac{PQ}{1714}$$

3.3.3 Other Components



In this example of a circuit utilizing sequence valves, when the left side of the valve is used such that both lines are straight, the left cylinder extends then the right one as the path to extend the left cylinder is directly connected while the flow control valve blocks the right. Vice versa on the other side the right cylinder retracts then the left one.

Accumulators



Accumulators act as the hydraulic equivalent of capacitors. Using air pressure they can store pressure created by a pump to provide pressure during intermittent outages or to dampen shocks in a system (like how a capacitor can be used to filter signals). The calculations of it are beyond what we do in this course.